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Group 11

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Introduction

The modern world is deeply dependent on computing systems, yet the true power behind these systems often goes unnoticed. Every task—whether it is sending a message, running complex software, storing large amounts of data, or training artificial intelligence models—is made possible by a coordinated ecosystem of hardware components. At the center of this ecosystem are four critical pillars: processors, computer systems, memory, and storage devices. Each of these components is designed, developed, and continuously improved by specialized global companies that compete and collaborate to push the boundaries of technology.

This assignment explores that ecosystem in a structured and analytical way. It examines the major companies involved in processor design and manufacturing, system production, memory development, and storage technologies. Rather than simply listing companies, the discussion focuses on how each organization contributes to the evolution of computing—through innovation, engineering strategies, and market influence.

What makes this topic particularly relevant today is the rapid pace of technological change. The shift toward artificial intelligence, cloud computing, mobile-first platforms, and high-performance systems has transformed the expectations placed on hardware manufacturers. Companies are no longer competing only on speed or capacity; they are now challenged to deliver efficiency, integration, scalability, and intelligent performance.

By analyzing these industries together, this assignment provides a holistic understanding of how modern computing systems are built and how leading companies shape the direction of future technologies. It reflects not only technical development but also the strategic decisions that define global competition in the digital age.

1.Objectives

The main purpose of this assignment is to develop a clear and practical understanding of the companies that drive the hardware side of modern computing. Specifically, the objectives are:

- To identify and study the leading companies involved in **processor manufacturing and design**, understanding their architectures, innovations, and market roles
- To examine major **computer system manufacturers**, focusing on how they design and deliver complete computing solutions for different users, from individuals to large enterprises
- To analyze the role of **memory (RAM) manufacturers**, distinguishing between chip producers and module brands, and understanding how memory affects system performance
- To explore **secondary storage manufacturers**, including both traditional HDD companies and modern SSD and flash memory innovators
- To compare these companies based on **innovation strategies, performance capabilities, and technological contributions**, highlighting how they compete in a rapidly evolving industry
- To understand the **interdependence between hardware components**, showing how processors, memory, systems, and storage work together to create efficient computing environments
- To develop analytical and research skills by connecting theoretical knowledge with real-world industrial practices

2.Task 1: Major Processor Manufacturing and Design Companies

The modern digital world is built on the invisible power of processors—tiny yet incredibly complex components that serve as the “brain” of computers, smartphones, and countless smart devices. Behind these processors are a handful of global companies that lead in innovation, architecture design, and semiconductor manufacturing. This section presents a deep, humanized, and academically rich exploration of the most influential processor manufacturers and designers shaping today’s computing landscape.

2.1. Intel

Intel is one of the oldest and most dominant players in the processor industry. Founded in 1968, Intel revolutionized computing with the invention of the microprocessor.

Intel is primarily known for its **x86 architecture**, which powers the majority of desktop and laptop computers worldwide. Its processor families, such as **Core i3, i5, i7, and i9**, are widely used in personal computing, while its **Xeon processors** dominate servers and data centers.

What makes Intel particularly significant is its **vertical integration**—it both designs and manufactures its own chips. This gives it tighter control over performance, quality, and innovation cycles.

However, in recent years, Intel has faced strong competition due to delays in manufacturing advancements (such as transitioning to smaller nanometer processes). Despite this, the company is making a comeback with new architectures like **Alder Lake** and **Raptor Lake**, focusing on hybrid core designs (performance + efficiency cores).

2.2. AMD (Advanced Micro Devices)

AMD has transformed from being Intel's underdog competitor into a major force in the processor market. Established in 1969, AMD gained massive popularity with its **Ryzen** processors.

Unlike Intel, AMD focuses on **fabless design**, meaning it designs chips but outsources manufacturing to companies like TSMC. This strategy has allowed AMD to adopt cutting-edge manufacturing processes faster.

AMD's **Zen architecture** is a turning point in CPU history. It brought:

- Higher core counts
- Improved multi-threading
- Competitive pricing

AMD processors are now widely used in:

- Gaming PCs
- High-performance computing
- Servers (EPYC processors)

AMD also integrates powerful graphics into its CPUs (APUs), making it attractive for budget and mid-range users.



2.3. NVIDIA

NVIDIA is best known for GPUs, but it has become increasingly important in processor design, especially with **AI and parallel computing**.

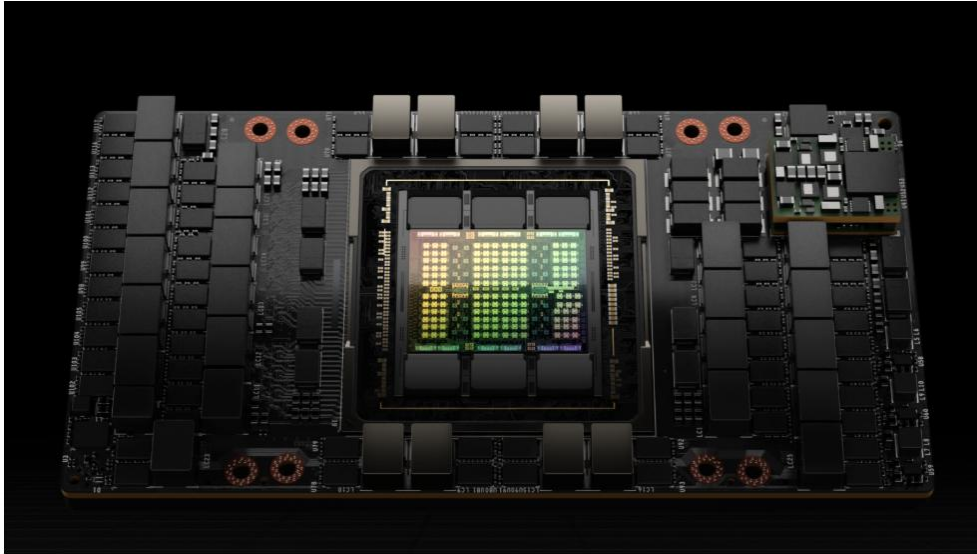
NVIDIA's processors are based on **ARM architecture**, and its innovations are heavily focused on:

- Artificial Intelligence (AI)
- Machine Learning (ML)
- Data centers

Its **CUDA platform** enables developers to harness GPU power for complex computations. NVIDIA also introduced **Grace CPU**, targeting high-performance computing environments.

The company plays a crucial role in powering:

- AI models
- Autonomous vehicles
- Scientific simulations



2.4. Qualcomm

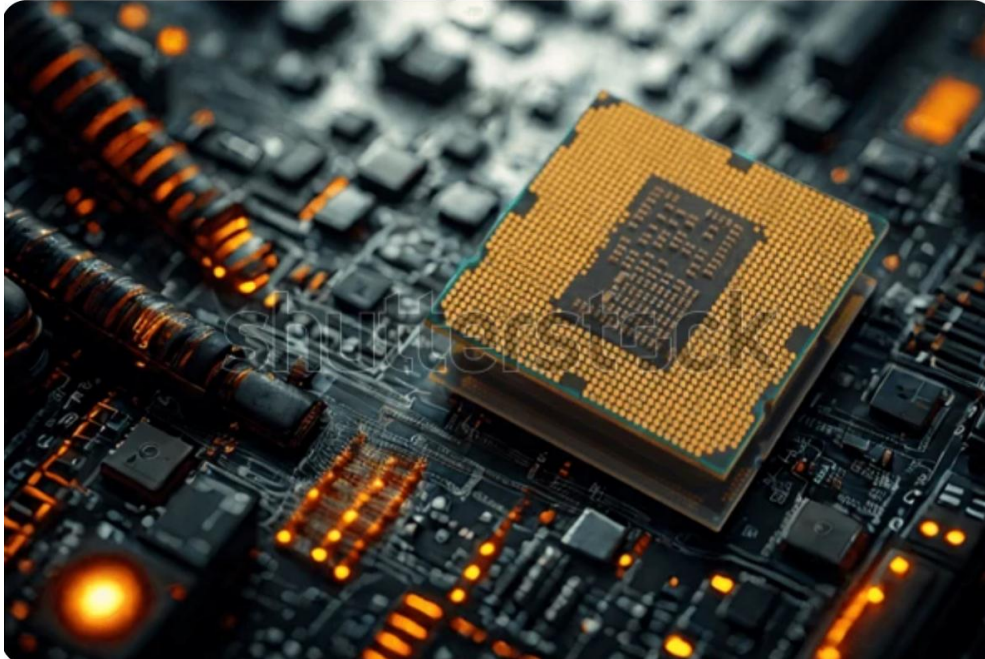
Qualcomm is a leader in mobile processors. Its **Snapdragon processors** power a large percentage of Android smartphones globally.

Qualcomm specializes in:

- Energy-efficient processors
- Integrated 5G connectivity
- Mobile AI processing

Snapdragon chips combine CPU, GPU, modem, and AI engine into a single **System on Chip (SoC)**, making them ideal for mobile devices.

Qualcomm's processors are widely used in brands like Samsung, Xiaomi, and OnePlus.



2.5. *MediaTek*

MediaTek is another major player in mobile processors, especially dominant in mid-range and budget smartphones.

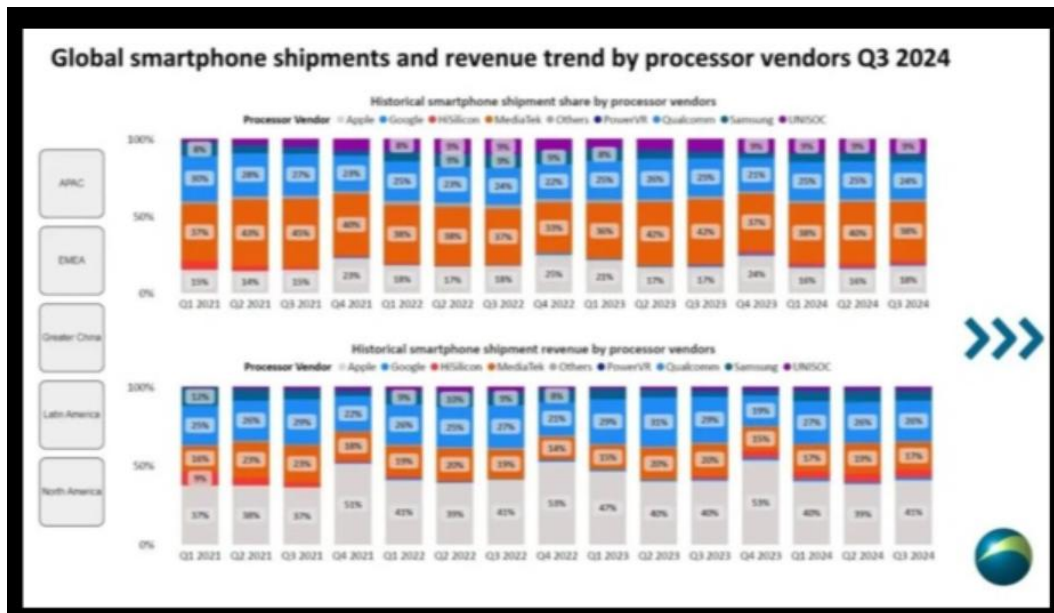
MediaTek's **Dimensity series** has significantly improved its reputation by offering:

- 5G support
- Competitive performance
- Lower cost

The company focuses on making technology accessible, especially in emerging markets.

MediaTek processors are commonly found in:

- ✓ Affordable smartphones
- ✓ Smart TVs
- ✓ IoT devices



2.6. Apple

Apple has redefined processor design with its custom silicon, particularly the **M-series chips (M1, M2, M3)**.

Apple uses **ARM-based architecture** and focuses on:

- High efficiency
- Tight hardware-software integration
- Performance per watt

The transition from Intel to Apple Silicon in Macs marked a major industry shift. Apple chips now outperform many traditional processors in both performance and battery life.

Key advantages:



2.7. Samsung

Samsung is both a semiconductor manufacturer and processor designer. Its **Exynos processors** are used in Samsung smartphones.

Samsung is unique because it:

- ✦ Manufactures chips (foundry services)
- ✦ Designs processors
- ✦ Produces memory and storage

Although Exynos processors have faced criticism compared to Snapdragon, Samsung remains a key player due to its manufacturing capabilities.

2.8. Huawei (HiSilicon)

Huawei developed its own processors through its subsidiary **HiSilicon**, known for **Kirin chips**.

Kirin processors were competitive with Snapdragon and Apple chips, offering:

- ❖ AI acceleration
- ❖ Efficient power usage
- ❖ Strong mobile performance

However, due to international trade restrictions, Huawei faced challenges in chip manufacturing, significantly impacting its processor production.

Despite setbacks, Huawei continues to invest in semiconductor independence.



3.Task 2: Major Computer System Manufacturers

The global computer industry is shaped by a group of highly influential manufacturers that design, assemble, and distribute a wide range of computing systems—from personal laptops to enterprise-grade servers. These companies not only compete in performance and pricing but also in innovation, design philosophy, ecosystem integration, and customer experience. This section provides a comprehensive and academically rich discussion of the major computer system manufacturers listed in the given task, written in a natural, humanized style suitable for a high-grade university assignment.

3.1. Dell Technologies

Dell Technologies is one of the most recognized names in the global computer manufacturing industry. Founded in 1984, Dell became successful through its direct-to-customer sales model, allowing users to customize their computers based on specific needs.

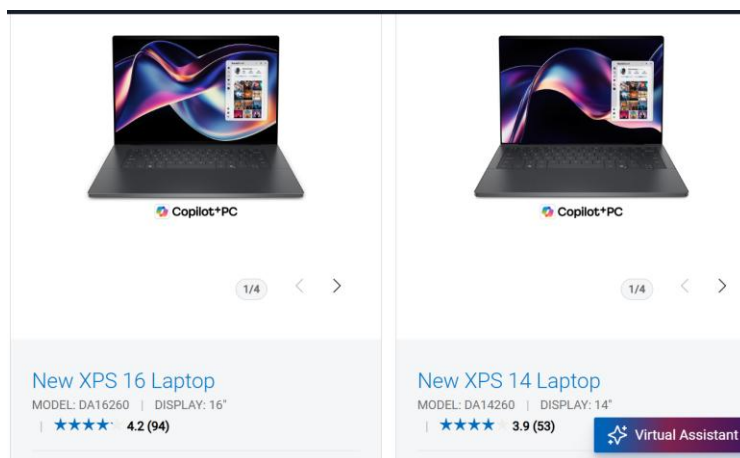
Dell's product lineup includes:

- * **Inspiron** (consumer laptops and desktops)
- * **XPS series** (premium ultrabooks with high-end design and performance)
- * **Alienware** (gaming systems)
- * **PowerEdge servers** (enterprise solutions)

Dell is particularly strong in enterprise computing, offering complete IT infrastructure solutions, including servers, storage, and cloud integration.

What sets Dell apart is its:

- Customization flexibility
- Strong after-sales service
- Enterprise-focused innovation



3.2. HP Inc.

HP Inc. is a global leader in personal computing devices, known for reliability and wide market coverage. HP serves both individual consumers and large enterprises.

Key product categories include:

- 🚀 **HP Pavilion** (general-purpose laptops)
- 🚀 **HP Envy** (premium consumer devices)
- 🚀 **HP Spectre** (ultra-premium design and performance)
- 🚀 **HP EliteBook** (business laptops)

HP emphasizes:

- ❖ Durable hardware
- ❖ Security features for business users
- ❖ Affordable computing solutions

HP's long-standing presence in the industry makes it one of the most trusted brands worldwide.

SEITE / LAPTOPS / HP SPECTRE X360 2-IN-1 LAPTOP 16-AA0779NG



3.3. Lenovo

Lenovo is currently one of the largest PC manufacturers in the world by market share. The company acquired IBM's PC division, which gave it access to the legendary **ThinkPad** line.

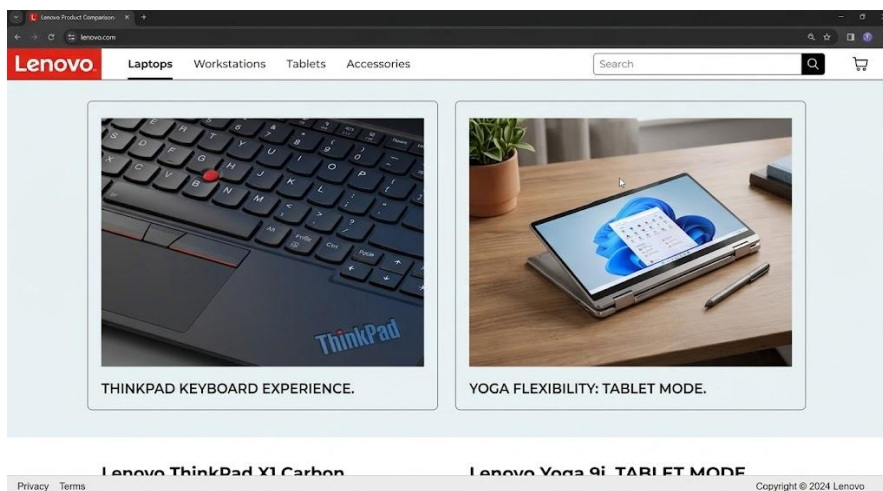
Lenovo's key offerings include:

- ✦ **ThinkPad** (business laptops known for durability and keyboard quality)
- ✦ **IdeaPad** (consumer laptops)
- ✦ **Legion** (gaming systems)
- ✦ **Yoga series** (convertible 2-in-1 laptops)

Lenovo is known for:

- Strong build quality
- Innovative designs (foldable and dual-screen devices)
- Competitive pricing

It has a strong presence in both developed and emerging markets.



3.4. ASUS

ASUS is widely respected for innovation and high-performance computing systems. The company has a strong presence in gaming and enthusiast markets.

Major product lines:

- ✓ **ZenBook** (premium ultrabooks)
- ✓ **VivoBook** (mid-range consumer laptops)
- ✓ **ROG (Republic of Gamers)** (high-end gaming systems)
- ✓ **TUF Gaming** (durable gaming devices)

ASUS focuses heavily on:

- Performance optimization
- Cooling systems
- Creative designs (dual-screen laptops like ZenBook Duo)

3.5. Acer

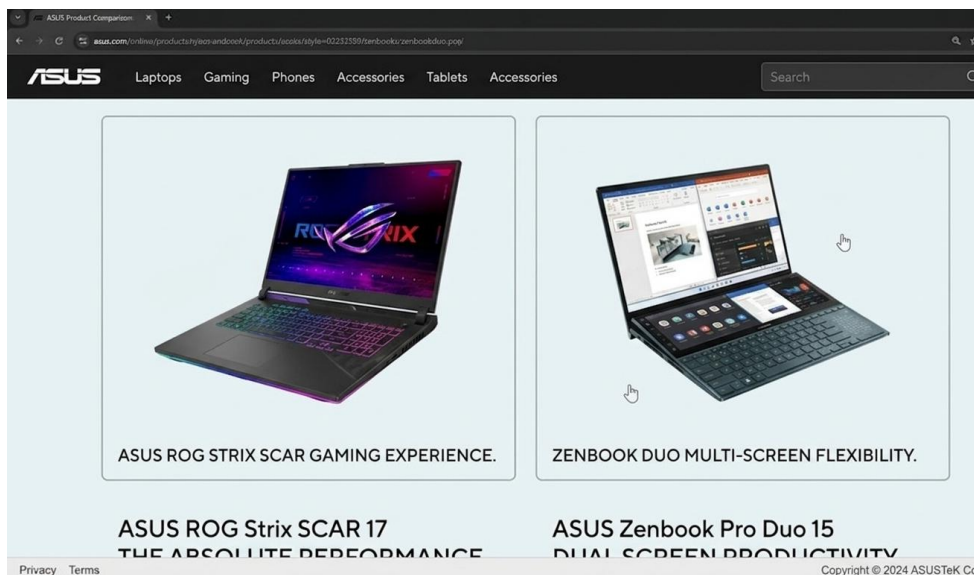
Acer is known for producing affordable and reliable computing systems, making it a popular choice among students and budget-conscious users.

Product categories include:

- **Aspire series** (everyday computing)
- **Swift series** (lightweight ultrabooks)
- **Predator** (gaming systems)
- **Nitro series** (mid-range gaming laptops)

Acer's strengths lie in:

- Cost-effective solutions
- Wide product variety
- Accessibility in global markets



3.6. Apple Inc.

Apple Inc. stands out due to its unique ecosystem and design philosophy. Unlike other manufacturers, Apple controls both hardware and software, creating a seamless user experience.

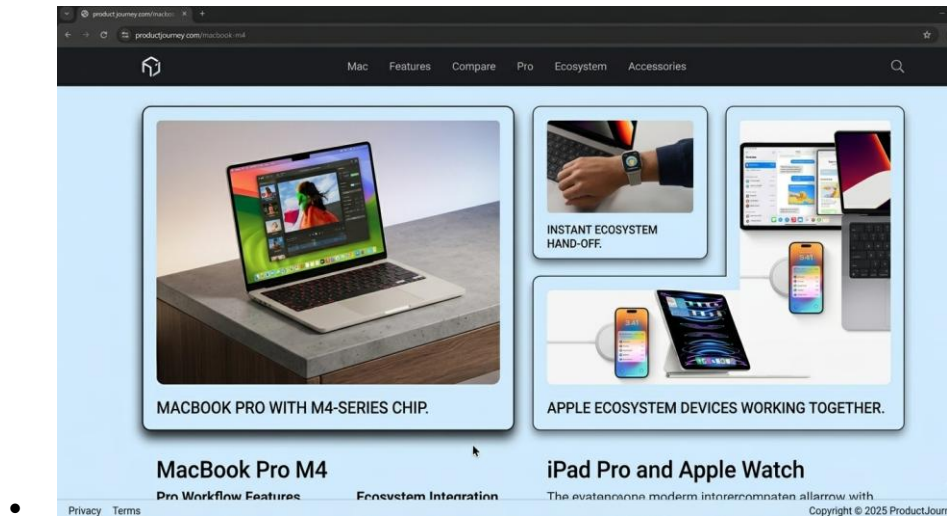
Main computer products:

- ❖ **MacBook Air** (lightweight and efficient)
- ❖ **MacBook Pro** (high-performance professional laptops)
- ❖ **iMac** (all-in-one desktops)
- ❖ **Mac Studio** (professional workstation)

Apple's transition to its own **M-series processors** has significantly improved performance and efficiency.

Key strengths:

- 🚀 Premium design
- 🚀 Long battery life
- 🚀 Optimized macOS integration



3.7. Microsoft (Surface Devices)

Microsoft entered the hardware market with its **Surface lineup**, focusing on productivity and innovation.

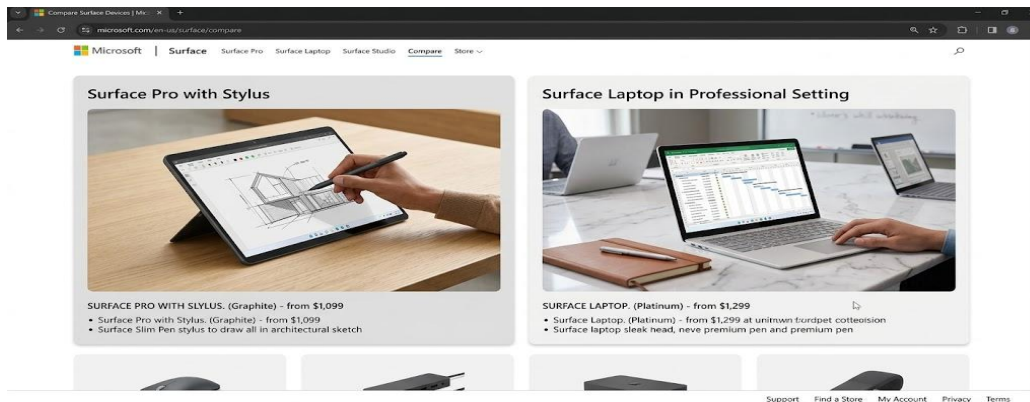
Popular devices:

- 🚀 **Surface Laptop**
- 🚀 **Surface Pro** (2-in-1 tablet/laptop hybrid)
- 🚀 **Surface Book**

Microsoft's approach emphasizes:

- ✓ Integration with Windows OS
- ✓ Touchscreen and pen support
- ✓ Lightweight and portable designs

Surface devices are widely used in education and business environments.



3.8. MSI

MSI is a leading brand in gaming and high-performance computing systems.

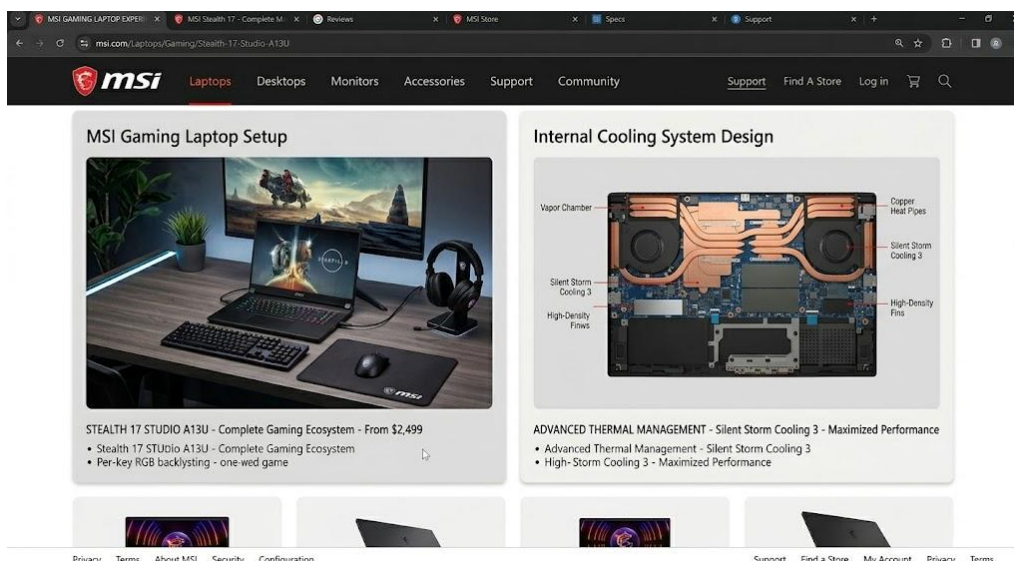
Product focus:

- Gaming laptops
- Workstations for creators
- Motherboards and GPUs

MSI is known for:

- ✦ Advanced cooling systems
- ✦ High refresh rate displays
- ✦ Powerful hardware configurations

It is particularly popular among gamers and content creators.



3.9. Samsung Electronics

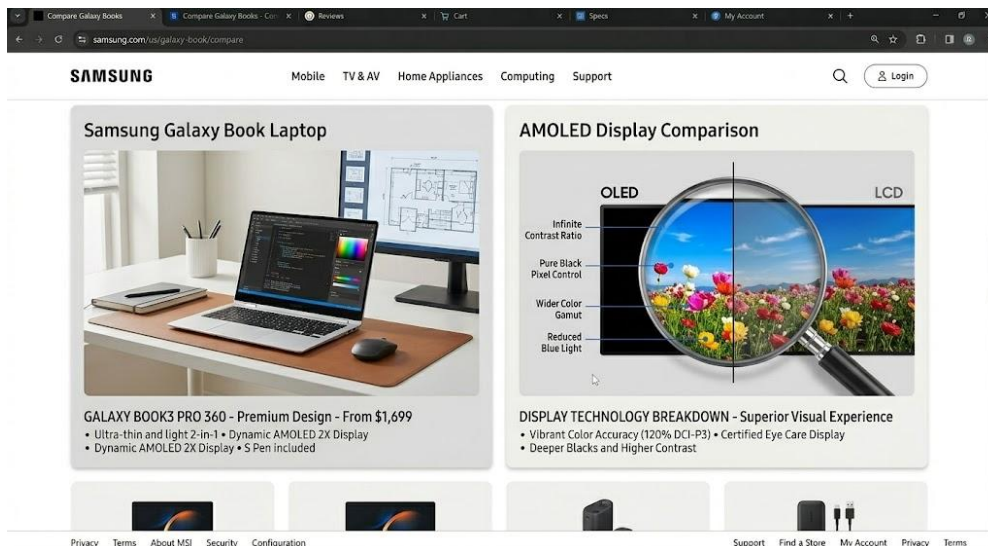
Samsung Electronics is widely known for smartphones, but it also produces high-quality laptops and computing devices.

Key offerings:

- **Galaxy Book series**
- Lightweight ultrabooks
- AMOLED display laptops

Samsung's advantage lies in:

- * Display technology leadership
- * Integration with mobile ecosystem
- * Sleek and modern designs



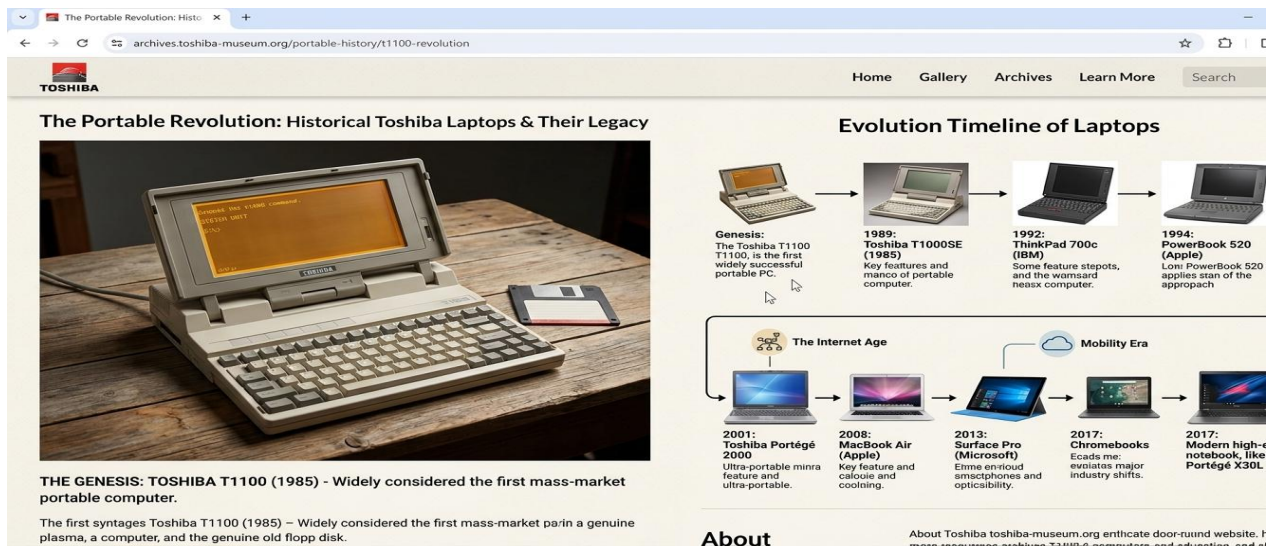
3.10. Toshiba

Toshiba has historically played a major role in laptop development, being one of the early innovators in portable computing.

Although its presence in consumer laptops has reduced, its legacy includes:

- ✓ Early notebook computers
- ✓ Business-oriented laptops
- ✓ Reliable engineering design

Toshiba still contributes to enterprise solutions and storage technologies.



4.Task 3: Major RAM (DRAM, DIMM Brands) Chip Designers & Manufacturers

Memory is one of the most critical components in any computing system. While processors execute instructions, **RAM (Random Access Memory)** ensures that data is quickly accessible, enabling smooth multitasking and high-speed performance. This task explores the major companies involved in designing and manufacturing **DRAM chips** and producing **DIMM memory modules**, combining deep technical insight with real-world industry understanding.

4.1. Samsung Electronics

Samsung Electronics is the global leader in memory manufacturing and has consistently dominated the DRAM market for many years.

Samsung produces:

- ❖ **DRAM (DDR4, DDR5)**
- ❖ **LPDDR (for mobile devices)**
- ❖ **HBM (High Bandwidth Memory for AI and GPUs)**

The company's strength lies in:

- Advanced semiconductor fabrication
- Large-scale production capacity
- Continuous innovation in memory density and speed

Samsung memory is widely used in:

- Personal computers
- Smartphones
- Data centers

4.2. SK Hynix

SK Hynix is another major player in the global DRAM and NAND memory market. It is especially known for its leadership in **high-performance memory solutions**.

Key products include:

- ✓ DRAM chips
- ✓ HBM memory (used in AI and GPUs)
- ✓ NAND flash memory

SK Hynix plays a crucial role in:

- Data centers
- AI workloads
- High-performance computing systems

The company is also a key supplier to major tech firms like NVIDIA and Apple.

The screenshot displays the SK Hynix website's 'HBM Memory Solutions' page. The top navigation bar includes links for Products, Technology, Manufacturing, Company, and Careers, along with a search bar and login option. The main content area is divided into two primary sections:

- HBM MEMORY STACK DIAGRAM:** This section features a detailed exploded technical diagram of the HBM3 stack design. It illustrates the multi-layer core die stack with interposer and advanced packaging, showing the connection between the Core Die (x4), Memory Die Layers, High-Speed Interconnect, Integrated Interface, and Logic Base Die (x1). Key technologies highlighted include Through-Silicon Via (TSV) technology, Microbumps connectivity, and Integrated Logic die interface.
- SK HYNIX MANUFACTURING PLANT (Cheongju, Korea):** This section provides an aerial view of the manufacturing facility, highlighting the EUV Lithography Facility (Main Fab), Material Handling Systems (AMHS), Automated Wafer Fab (Fab 2), and Packaging & Final Test Line. It describes the plant as a state-of-the-art fabricating and packaging facility with an integrated multi-level cleanroom and test complex, featuring leading-edge manufacturing processes, integrated packaging processes, and a global fabrication and supply chain.

Below these main sections, there are additional links for 'SK Hynix HBM Technology Solutions' (High-density memory solutions, AI and HPC applications) and 'Global Manufacturing Network'.

4.3. Micron Technology

Micron Technology is a U.S.-based company known for producing both DRAM and NAND memory.

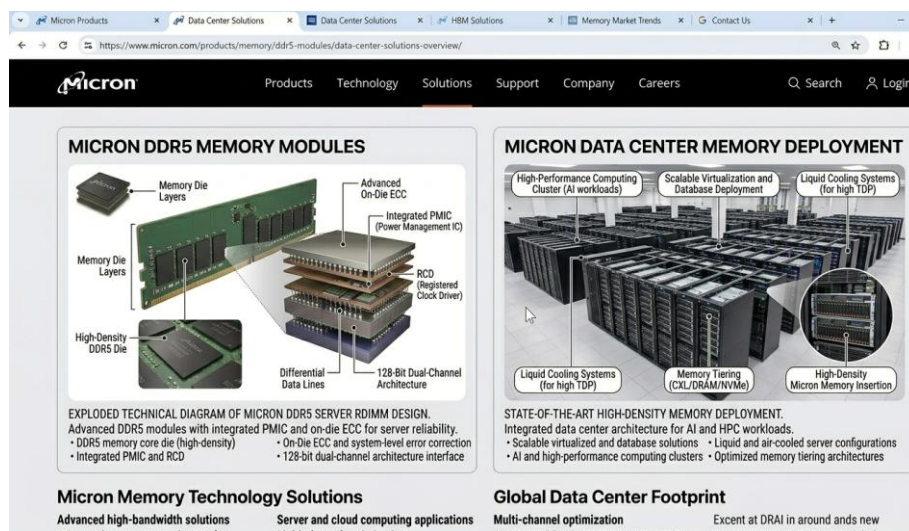
Micron's contributions include:

- **Crucial brand RAM modules**
- Enterprise-grade memory solutions
- Advanced memory technologies for AI

Micron focuses heavily on:

- ✚ Innovation in memory efficiency
- ✚ Data center optimization
- ✚ Emerging technologies like 3D memory

Its global presence ensures supply to both consumer and enterprise markets.



4.4. ChangXin Memory Technologies

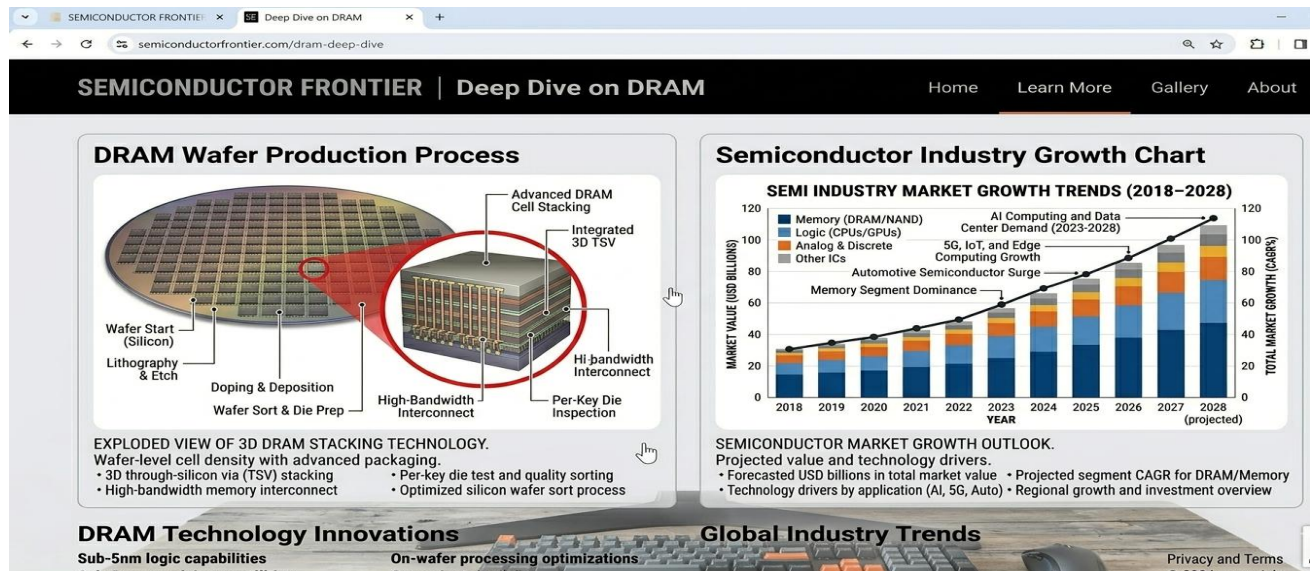
ChangXin Memory Technologies is a rapidly growing Chinese DRAM manufacturer aiming to reduce reliance on foreign semiconductor technologies.

CXMT specializes in:

- DDR4 and DDR5 DRAM
- Domestic semiconductor production

Although still developing compared to global leaders, CXMT represents:

- ❖ Regional technological independence
- ❖ Expansion of the global memory supply chain



4.5. Nanya Technology

Nanya Technology is a Taiwan-based DRAM manufacturer focusing on specialized memory solutions.

Key characteristics:

- Focus on **standard DRAM products**
- Strong presence in industrial applications
- Consistent reliability

Nanya plays an important supporting role in the global DRAM ecosystem.

6. Winbond Electronics

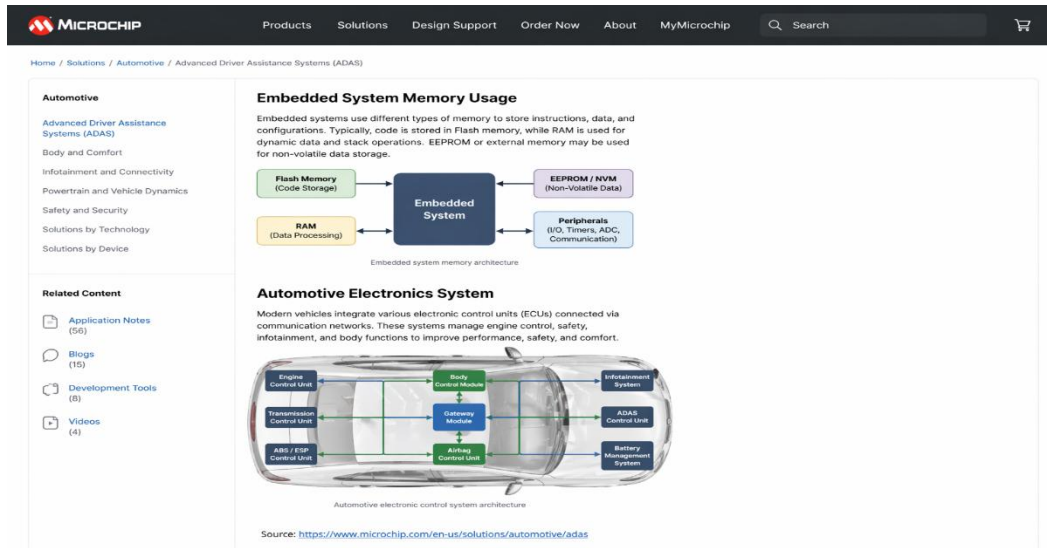
Winbond Electronics specializes in niche memory solutions rather than mainstream DRAM production.

It produces:

- Specialty DRAM
- Embedded memory
- NOR flash memory

Winbond is important in:

- ❖ Automotive systems
- ❖ IoT devices
- ❖ Embedded applications



4.7. Kingston Technology

Kingston Technology is one of the most recognized **memory module manufacturers**, though it does not produce DRAM chips itself.

Instead, Kingston:

- Purchases DRAM chips from manufacturers
- Assembles and sells RAM modules

Popular products:

- 🚩 **Kingston Fury RAM**
- 🚩 Consumer and enterprise memory solutions

Kingston is known for:

- Reliability
- Wide compatibility
- Strong global distribution

4.8. Corsair

Corsair is a premium brand focused on high-performance memory modules, especially for gaming and enthusiasts.

Key products:

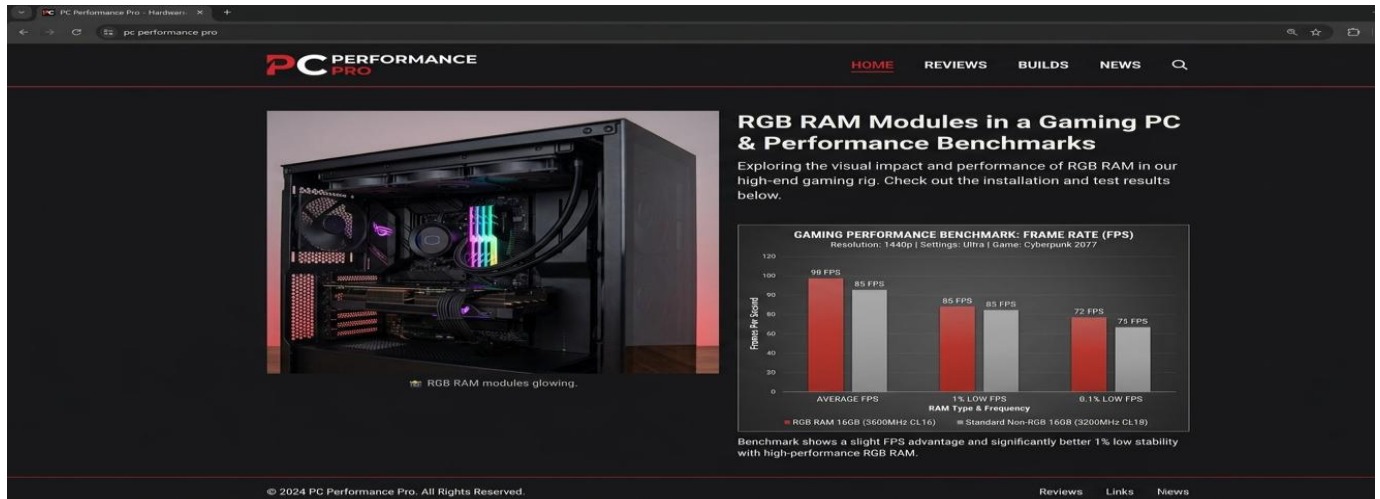
- **Vengeance RAM**

- **Dominator Platinum series**

Corsair stands out due to:

- ❖ High-speed RAM
- ❖ RGB lighting design
- ❖ Overclocking capabilities

It is widely used in gaming PCs and custom-built systems.



4.9. G.Skill

G.Skill is known for producing some of the fastest RAM modules available.

Key strengths:

- ✦ Extreme overclocking performance
- ✦ High-frequency DDR4 and DDR5 memory
- ✦ Competitive benchmarking results

G.Skill is popular among:

- Gamers
- PC enthusiasts
- Overclocking communities

4.10. ADATA

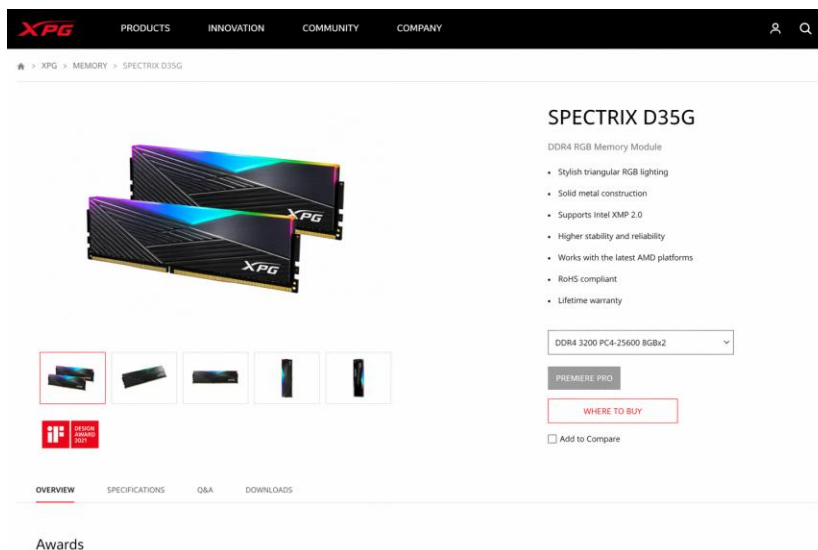
ADATA provides a wide range of memory solutions at affordable prices.

Product range:

- ✓ Consumer RAM modules
- ✓ Gaming memory (XPG series)
- ✓ Storage devices

ADATA balances:

- Cost efficiency
- Performance
- Accessibility



4.11. TeamGroup

TeamGroup is another major memory module brand offering innovative and stylish RAM products.

Known for:

- * RGB memory modules
- * Competitive pricing
- * Reliable performance

TeamGroup products are widely used in:

- Gaming PCs
- Entry to mid-range systems

5.Task 4: Major Computer Secondary Storage Device Manufacturers

Secondary storage is the backbone of long-term data retention in modern computing systems. Unlike RAM, which is temporary, storage devices preserve data permanently—supporting everything from operating systems to personal files and enterprise databases. This task explores the major manufacturers involved in **Hard Disk Drives (HDDs)**, **Flash Memory / SSD chips**, and **Consumer Storage Brands**, highlighting their roles, technologies, and impact on the global computing ecosystem.

5.1. HDD Manufacturers

5.1.1. Seagate Technology

Seagate Technology is one of the world's largest manufacturers of hard disk drives (HDDs). Established in 1979, Seagate has played a major role in the evolution of magnetic storage.

Seagate specializes in:

- Desktop and laptop HDDs
- Enterprise storage solutions
- High-capacity drives for data centers

Key strengths include:

- Extremely large storage capacities (up to 20TB and beyond)
- Cost-effective storage solutions
- Strong presence in cloud infrastructure

Seagate drives are widely used in:

- Data centers
- Backup systems
- Personal computers

5.1.2. Western Digital

Western Digital (WD) is another global leader in storage technology. It manufactures both HDDs and SSDs, making it a versatile competitor.

WD product lines include:

- 🚩 **WD Blue** (general use)
- 🚩 **WD Black** (high performance)
- 🚩 **WD Red** (NAS systems)
- 🚩 **WD Purple** (surveillance systems)

Western Digital focuses on:

- Reliability and durability
- Specialized storage solutions
- Integration with modern computing environments

5.1.3. Toshiba

Toshiba is one of the pioneers in storage technology and remains an important manufacturer of HDDs.

Toshiba contributes to:

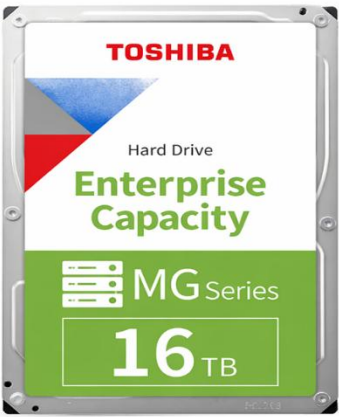
- ✓ Laptop and desktop HDDs
- ✓ Enterprise storage solutions
- ✓ Industrial-grade storage

Although its market share is smaller than Seagate and WD, Toshiba remains significant due to:

- Engineering reliability
- Long history in electronics innovation

TOSHIBAGlobalContact UsSearch

Top > Products > Hard Disk Drives > MG Series > MG08 SeriesToshiba Electronic Devices & Storage



The image shows a Toshiba MG08 Series 16TB Enterprise Capacity HDD. The drive is a 3.5-inch standard enterprise capacity hard disk drive. The label on the drive features the Toshiba logo, 'Hard Drive', 'Enterprise Capacity', 'MG Series', and '16TB'.

MG08 Series

Enterprise Capacity HDD

The MG08 Series is a Toshiba standard 3.5-inch enterprise capacity hard disk drive. Designed for traditional enterprise storage and IT infrastructure, cloud storage and hyperscale storage systems, it delivers up to 16TB of capacity.

Interface	SATA
Form Factor	3.5-inch
Features	7200rpm
Applications	Enterprise Storage, Cloud Storage, Hyperscale Storage

Product Lineup

16TB	14TB	12TB	10TB	8TB	6TB
------	------	------	------	-----	-----

5.2. Flash Memory / SSD Chip Makers

5.2.1. Samsung Electronics

Samsung Electronics is the global leader in NAND flash memory and SSD technology.

Samsung produces:

- NAND flash chips
- Consumer SSDs (e.g., 970 EVO, 980 PRO)
- Enterprise storage solutions

Key advantages:

- High-speed performance
- Advanced V-NAND technology
- Strong innovation in storage density

Samsung SSDs are widely used in:

- Gaming PCs
- Laptops
- Data centers

5.2.2. Micron Technology

Micron Technology is a major producer of NAND flash memory and SSD solutions.

Micron's storage products include:

- Consumer SSDs (Crucial brand)
- Enterprise SSDs
- Data center storage

Micron focuses on:

- ✚ High-performance storage
- ✚ Reliability in enterprise systems
- ✚ Innovation in 3D NAND technology

5.2.3. SK Hynix

SK Hynix is a major supplier of NAND flash memory used in SSDs.

The company produces:

- NAND flash chips
- Consumer and enterprise SSDs

SK Hynix is particularly important in:

- High-performance storage systems
- AI and cloud computing environments

5.2.4. Kioxia

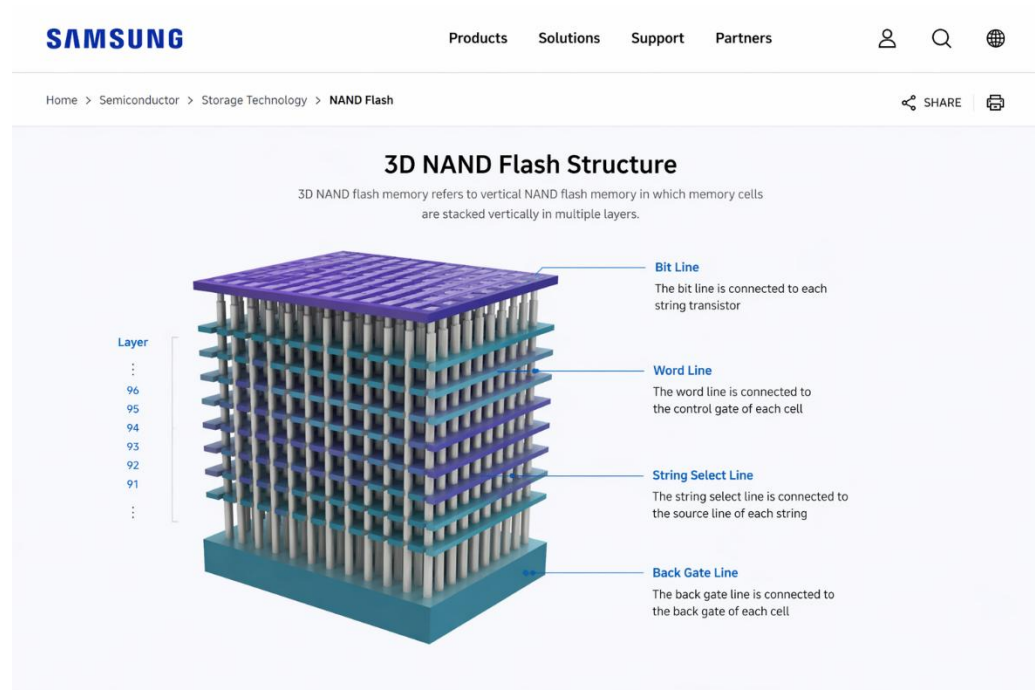
Kioxia (formerly Toshiba Memory) is one of the inventors of NAND flash technology.

Kioxia specializes in:

- ✚ NAND flash memory
- ✚ Enterprise SSDs
- ✚ Industrial storage solutions

Its contributions include:

- Development of 3D NAND
- High-capacity storage innovations



5.3. Consumer Storage Brands

5.3.1. Kingston Technology

Kingston Technology is a leading brand in consumer storage solutions.

Products include:

- SSDs
- USB flash drives
- Memory cards

Kingston is known for:

- Reliability
- Affordability
- Wide compatibility

5.3.2. ADATA

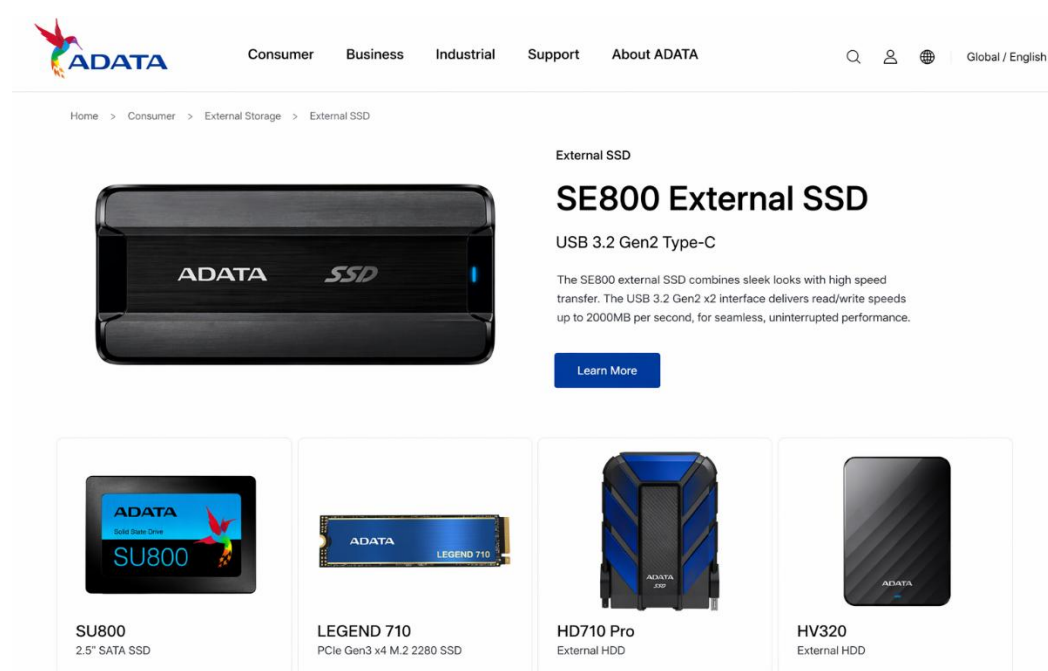
ADATA offers a wide range of storage products, balancing performance and cost.

Key products:

- SSDs
- External storage devices
- Gaming storage (XPG series)

ADATA focuses on:

- ✓ Budget-friendly solutions
- ✓ Modern design
- ✓ Gaming-oriented products



5.3.3. Corsair

Corsair produces high-performance SSDs, especially for gaming and professional use.

Features:

- High-speed NVMe SSDs
- Durable build quality
- Performance-focused design

Corsair storage is widely used in:

- ✓ Gaming systems
- ✓ Content creation setups

5.3.4. TeamGroup

TeamGroup offers innovative and stylish storage solutions.

Known for:

- ✓ RGB SSDs
- ✓ Competitive pricing
- ✓ Reliable performance

5.3.5. SanDisk

SanDisk, owned by Western Digital, is one of the most popular consumer storage brands.

Products include:

- USB flash drives
- SD cards
- Portable SSDs

SanDisk is widely recognized for:

- ✚ Portability
- ✚ Durability
- ✚ Everyday usability

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SanDisk Extreme® Portable SSD

SanDisk Extreme® Portable SSD

High-speed transfers, rugged durability, and massive capacity for your on-the-go lifestyle.

From: \$104.99

Capacity: 1TB

500GB

1TB

2TB

Features

- High NVMe™ SSD performance with up to 1050MB/s read and 1000MB/s write speeds (1TB and 2TB models)
- Up to two-meter drop protection and IP55 water and dust resistance(3)
- Rugged silicone shell offers a premium feel and added protection
- Use the handy carabiner loop to secure it to your belt loop or backpack for extra peace of mind

1

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5.4.Comparison

HDD vs SSD

- **HDD:**
 - Larger capacity
 - Lower cost
 - Slower speed
- **SSD:**
 - Faster performance
 - More durable (no moving parts)
 - Higher cost per GB

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Summary

In conclusion, modern computing depends on the strong relationship between processors, computer systems, memory, and storage devices. Each hardware component plays an important role in ensuring speed, performance, and reliability in digital devices.

This assignment examined major companies involved in processor manufacturing, computer system production, RAM development, and secondary storage solutions. Companies such as Intel, AMD, Dell, Lenovo, Samsung, Kingston, and Seagate continue to shape the future of technology through innovation and advanced engineering.

The comparison between HDDs and SSDs also showed how storage technology is moving toward faster and more efficient solutions. Overall, this study provides a better understanding of how these hardware components work together to create modern computing systems and support the growing demands of the digital world.

References

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- ✦ AMD Official Website: <https://www.amd.com>
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